

TELCO-CLOUD ENGINEERING

The Telco-Cloud Engineering Track focuses on next-generation wireless networks. It seamlessly blends advanced telecom infrastructure with cutting-edge cloud technologies, machine learning, and autonomous, agentic AI systems. Learners dive deep into modern cloud-native infrastructures. They master the deployment, optimization, and automated orchestration of 5G Cloud RAN (CRAN) and Open RAN (ORAN) architectures.

Students work on real-world production challenges in the capstone graduation projects, which are designed, sponsored, and directly mentored by top-tier global telecom leaders.

An intensive soft skills package is also delivered so that graduates can deeply understand how to manage projects and build teams.

JOB PROFILES

**Cloud-RAN Engineer · Open-RAN Engineer ·
Core Network Engineer · 5G MLOps Engineer · Telco-Cloud Engineer**



TOTAL HOURS

1,333hrs

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Minimum grade required



CERTIFICATIONS

2


Where graduates work

WE

e& Egypt

Vodafone

_VOIS

Orange

OpenValley

DigisSquared

SyChield

SiVision

Huawei

Nokia

Ericsson



Certifications earned

AWS — Certified CloudOps Engineer – Associate

HCIA-5G-RAN — Huawei Certified ICT Associate



Selection process

01 IQ & English Exam

Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam

Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview

Interview with the technical committee.

04 Interpersonal Interview

Candidates who pass the technical interview are invited to this final stage.



Who may apply

Open to graduates in Communications Engineering, with a minimum grade of Good or higher.

Program content – 1,333 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Fundamental Courses	327 hrs	25%	Operating Systems Fundamentals, Cisco Network Associate, Introduction to Programming, Object oriented programming , Data Structures and Algorithms, Database Fundamentals, Linux Systems Fundamentals, Bash Shell Script, Python Programming , Introduction to Generative AI and Prompt Engineering, AI-Assisted Coding and AI Coding Agents, Advanced Gen-AI Programing: RAG and Agentic Software
● Telecommunication Core	315 hrs	24%	Fundamentals of Wireless Communications, Wireless Mobile Networks, LTE And LTE Advanced, RAN Optimization Fundamentals, RAN Automation Fundamentals , Cloud RAN Architecture , Open RAN Architecture , 5G and Beyond , 5G RAN and Edge Computing , 5G RAN Optimization
● Telco-Cloud Core	300 hrs	23%	Core Networks Fundamentals , AI/ML Fundamentals For 5G , 5G Core Network, 5G MLOps, Telco-Cloud Fundamentals , Telco-Cloud Technologies , Telco-Cloud Monitoring and Reporting , Telco-Cloud Analytics using AI, Digital Twins for Communication (Workshop)
● Soft Skills	91 hrs	7%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor
● Graduation Project	300 hrs	23%	

Hands-on labs are conducted in-person. Online components cover theory and self-study modules on MaharaTech.

Student Deliverable

Every graduate must obtain an international certification relevant to their specialisation and complete an industry-related graduation project.

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

🔗 **Programming Fundamentals**
<https://aws.amazon.com/certification/certified-cloudops-engineer-associate/>

🔗 **Python Fundamentals**
<https://maharatech.gov.eg/course/view.php?id=43>

🔗 **Linux Fundamentals**
<https://maharatech.gov.eg/course/view.php?id=2155>

🔗 **Mobile Networks Fundamentals**
<https://www.tutorialspoint.com/gsm/index.htm>

🔗 **Cloud and Virtualization Concepts**
 ITI Values & Mission

🔗 **ITI Values & Mission**
<https://iti.gov.eg/about-us>

APPLY AT

<https://iti.gov.eg>

Applications open each intake period. Check the website for current dates and deadlines.